Does Body Mass Index Predict Blood Pressure? Evidence from NHANES

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This notebook loads the NHANES data for our STAT203 project and prepares the datasets for later cleaning, merging, and regression analysis.

1. Load Required Packages

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(haven)
library(ggplot2)
library(stargazer)
```

```
##
## Please cite as:
##
##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

2. Load NHANES Data

NHANES stores different types of information in separate files. For this project, we will use the demographics file, the body measures file, and the blood pressure file. Each dataset contains information on the same individuals, and later we will merge them using the participant identifier `SEQN`.

```r
data_path <- "../data/external/"

demo <- read_xpt(file.path(data_path, "DEMO_J.xpt"))
bmx  <- read_xpt(file.path(data_path, "BMX_J.xpt"))
bpx  <- read_xpt(file.path(data_path, "BPX_J.xpt"))
smq  <- read_xpt(file.path(data_path, "SMQ_J.xpt"))
```

3. Inspect the Data

At this stage, the goal is simply to confirm that the files loaded correctly and to get familiar with the structure of each dataset.

```r
head(demo)
```

```
## # A tibble: 6 x 46
##    SEQN SDDSRVYR RIDSTATR RIAGENDR RIDAGEYR RIDAGEMN RIDRETH1 RIDRETH3 RIDEXMON
##   <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 93703       10        2        2        2       NA        5        6        2
## 2 93704       10        2        1        2       NA        3        3        1
## 3 93705       10        2        2       66       NA        4        4        2
## 4 93706       10        2        1       18       NA        5        6        2
## 5 93707       10        2        1       13       NA        5        7        2
## 6 93708       10        2        2       66       NA        5        6        2
## # i 37 more variables: RIDEXAGM <dbl>, DMQMILIZ <dbl>, DMQADFC <dbl>,
## #   DMDBORN4 <dbl>, DMDCITZN <dbl>, DMDYRSUS <dbl>, DMDEDUC3 <dbl>,
## #   DMDEDUC2 <dbl>, DMDMARTL <dbl>, RIDEXPRG <dbl>, SIALANG <dbl>,
## #   SIAPROXY <dbl>, SIAINTRP <dbl>, FIALANG <dbl>, FIAPROXY <dbl>,
## #   FIAINTRP <dbl>, MIALANG <dbl>, MIAPROXY <dbl>, MIAINTRP <dbl>,
## #   AIALANGA <dbl>, DMDHHSIZ <dbl>, DMDFMSIZ <dbl>, DMDHHSZA <dbl>,
## #   DMDHHSZB <dbl>, DMDHHSZE <dbl>, DMDHRGND <dbl>, DMDHRAGZ <dbl>, ...
```

```r
head(bmx)
```

```
## # A tibble: 6 x 21
##    SEQN BMDSTATS BMXWT BMIWT BMXRECUM BMIRECUM BMXHEAD BMIHEAD BMXHT BMIHT
##   <dbl>    <dbl> <dbl> <dbl>    <dbl>    <dbl>   <dbl>   <dbl> <dbl> <dbl>
## 1 93703        1  13.7     3     89.6       NA      NA      NA  88.6    NA
## 2 93704        1  13.9    NA     95         NA      NA      NA  94.2    NA
## 3 93705        1  79.5    NA     NA         NA      NA      NA 158.     NA
## 4 93706        1  66.3    NA     NA         NA      NA      NA 176.     NA
## 5 93707        1  45.4    NA     NA         NA      NA      NA 158.     NA
## 6 93708        1  53.5    NA     NA         NA      NA      NA 150.     NA
## # i 11 more variables: BMXBMI <dbl>, BMXLEG <dbl>, BMILEG <dbl>, BMXARML <dbl>,
## #   BMIARML <dbl>, BMXARMC <dbl>, BMIARMC <dbl>, BMXWAIST <dbl>,
## #   BMIWAIST <dbl>, BMXHIP <dbl>, BMIHIP <dbl>
```

```r
head(bpx)
```

```
## # A tibble: 6 x 21
##    SEQN PEASCCT1 BPXCHR BPAARM BPACSZ BPXPLS BPXPULS BPXPTY BPXML1 BPXSY1 BPXDI1
```

```
##   <dbl>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>   <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
## 1 93703       NA    120     NA     NA     NA       1     NA     NA     NA     NA
## 2 93704       NA    114     NA     NA     NA       1     NA     NA     NA     NA
## 3 93705       NA     NA      1      4     52       1      1    220     NA     NA
## 4 93706       NA     NA      1      3     82       1      1    140    112     74
## 5 93707       NA     NA      1      2    100       1      1    140    128     38
## 6 93708       NA     NA      1      3     68       1      1    170     NA     NA
## # i 10 more variables: BPAEN1 <dbl>, BPXSY2 <dbl>, BPXDI2 <dbl>, BPAEN2 <dbl>,
## #   BPXSY3 <dbl>, BPXDI3 <dbl>, BPAEN3 <dbl>, BPXSY4 <dbl>, BPXDI4 <dbl>,
## #   BPAEN4 <dbl>
```

```r
head(smq)
```

```
## # A tibble: 6 x 37
##    SEQN SMQ020 SMD030 SMQ040 SMQ050Q SMQ050U SMD057 SMQ078 SMD641 SMD650 SMD093
##   <dbl>  <dbl>  <dbl>  <dbl>   <dbl>   <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
## 1 93705      1     16      3      30       4      5     NA     NA     NA     NA
## 2 93706      2     NA     NA      NA      NA     NA     NA     NA     NA     NA
## 3 93707     NA     NA     NA      NA      NA     NA     NA     NA     NA     NA
## 4 93708      2     NA     NA      NA      NA     NA     NA     NA     NA     NA
## 5 93709      1     15      1      NA      NA     NA      1     30      5      1
## 6 93711      2     NA     NA      NA      NA     NA     NA     NA     NA     NA
## # i 26 more variables: SMDUPCA <chr>, SMD100BR <chr>, SMD100FL <dbl>,
## #   SMD100MN <dbl>, SMD100LN <dbl>, SMD100TR <dbl>, SMD100NI <dbl>,
## #   SMD100CO <dbl>, SMQ621 <dbl>, SMD630 <dbl>, SMQ661 <dbl>, SMQ665A <dbl>,
## #   SMQ665B <dbl>, SMQ665C <dbl>, SMQ665D <dbl>, SMQ670 <dbl>, SMQ848 <dbl>,
## #   SMQ852Q <dbl>, SMQ852U <dbl>, SMQ890 <dbl>, SMQ895 <dbl>, SMQ900 <dbl>,
## #   SMQ905 <dbl>, SMQ910 <dbl>, SMQ915 <dbl>, SMAQUEX2 <dbl>
```

```r
cat("Demographic Data Columns...\n\n")
```

```
## Demographic Data Columns...
```

```r
colnames(demo)
```

```
##  [1] "SEQN"     "SDDSRVYR" "RIDSTATR" "RIAGENDR" "RIDAGEYR" "RIDAGEMN"
##  [7] "RIDRETH1" "RIDRETH3" "RIDEXMON" "RIDEXAGM" "DMQMILIZ" "DMQADFC"
## [13] "DMDBORN4" "DMDCITZN" "DMDYRSUS" "DMDEDUC3" "DMDEDUC2" "DMDMARTL"
## [19] "RIDEXPRG" "SIALANG"  "SIAPROXY" "SIAINTRP" "FIALANG"  "FIAPROXY"
## [25] "FIAINTRP" "MIALANG"  "MIAPROXY" "MIAINTRP" "AIALANGA" "DMDHHSIZ"
## [31] "DMDFMSIZ" "DMDHHSZA" "DMDHHSZB" "DMDHHSZE" "DMDHRGND" "DMDHRAGZ"
## [37] "DMDHREDZ" "DMDHRMAZ" "DMDHSEDZ" "WTINT2YR" "WTMEC2YR" "SDMVPSU"
## [43] "SDMVSTRA" "INDHHIN2" "INDFMIN2" "INDFMPIR"
```

```r
cat("\n")
```

```r
cat("Body Measure Data Columns...\n\n")
```

```
## Body Measure Data Columns...
```

```r
colnames(bmx)
```

```
##  [1] "SEQN"     "BMDSTATS" "BMXWT"    "BMIWT"    "BMXRECUM" "BMIRECUM"
##  [7] "BMXHEAD"  "BMIHEAD"  "BMXHT"    "BMIHT"    "BMXBMI"   "BMXLEG"
## [13] "BMILEG"   "BMXARML"  "BMIARML"  "BMXARMC"  "BMIARMC"  "BMXWAIST"
## [19] "BMIWAIST" "BMXHIP"   "BMIHIP"
```

```r
cat("\n")
```

```r
cat("Blood Pressure Data Columns...\n\n")
```

```
## Blood Pressure Data Columns...
```

```r
colnames(bpx)
```

```
##  [1] "SEQN"     "PEASCCT1" "BPXCHR"   "BPAARM"   "BPACSZ"   "BPXPLS"
##  [7] "BPXPULS"  "BPXPTY"   "BPXML1"   "BPXSY1"   "BPXDI1"   "BPAEN1"
## [13] "BPXSY2"   "BPXDI2"   "BPAEN2"   "BPXSY3"   "BPXDI3"   "BPAEN3"
## [19] "BPXSY4"   "BPXDI4"   "BPAEN4"
```

```r
cat("\n")
```

```r
cat("Smoking Questioner Data Columns...\n\n")
```

```
## Smoking Questioner Data Columns...
```

```r
colnames(smq)
```

```
##  [1] "SEQN"     "SMQ020"   "SMD030"   "SMQ040"   "SMQ050Q"  "SMQ050U"
##  [7] "SMD057"   "SMQ078"   "SMD641"   "SMD650"   "SMD093"   "SMDUPCA"
## [13] "SMD100BR" "SMD100FL" "SMD100MN" "SMD100LN" "SMD100TR" "SMD100NI"
## [19] "SMD100CO" "SMQ621"   "SMD630"   "SMQ661"   "SMQ665A"  "SMQ665B"
## [25] "SMQ665C"  "SMQ665D"  "SMQ670"   "SMQ848"   "SMQ852Q"  "SMQ852U"
## [31] "SMQ890"   "SMQ895"   "SMQ900"   "SMQ905"   "SMQ910"   "SMQ915"
## [37] "SMAQUEX2"
```

```r
demo_small <- demo %>%
  select(SEQN, RIDAGEYR)

bmx_small <- bmx %>%
  select(SEQN, BMXBMI)

bpx_small <- bpx %>%
  select(SEQN, BPXSY1)

smq_small <- smq %>%
  select(SEQN, SMQ020)
```

```r
df <- demo_small %>%
  inner_join(bmx_small, by = "SEQN") %>%
  inner_join(bpx_small, by = "SEQN") %>%
  inner_join(smq_small, by = "SEQN")
```

```r
df
```

```
## # A tibble: 6,401 x 5
##     SEQN RIDAGEYR BMXBMI BPXSY1 SMQ020
##    <dbl>    <dbl>  <dbl>  <dbl>  <dbl>
##  1 93705       66   31.7     NA      1
##  2 93706       18   21.5    112      2
##  3 93707       13   18.1    128     NA
##  4 93708       66   23.7     NA      2
##  5 93709       75   38.9    120      1
##  6 93711       56   21.3    108      2
##  7 93712       18   19.7    112      1
##  8 93713       67   23.5    104      1
##  9 93714       54   39.9     NA      1
## 10 93715       71   22.5    112      1
## # i 6,391 more rows
```

```r
unique(df$SMQ020)
```

```
## [1]  1  2 NA
```

```r
df_clean <- df %>%
  drop_na() %>%
  mutate(
    smoking = ifelse(SMQ020 == 1, 1,
                     ifelse(SMQ020 == 2, 0, NA))
  ) %>%
  drop_na(smoking) %>%
  select(SEQN, RIDAGEYR, BMXBMI, BPXSY1, smoking)
df_clean
```

```
## # A tibble: 4,833 x 5
##     SEQN RIDAGEYR BMXBMI BPXSY1 smoking
##    <dbl>    <dbl>  <dbl>  <dbl>   <dbl>
##  1 93706       18   21.5    112       0
##  2 93709       75   38.9    120       1
##  3 93711       56   21.3    108       0
##  4 93712       18   19.7    112       1
##  5 93713       67   23.5    104       1
##  6 93715       71   22.5    112       1
##  7 93716       61   30.7    120       1
##  8 93717       22   24.5    116       1
##  9 93718       45   22      128       0
## 10 93721       60   35.9    132       0
## # i 4,823 more rows
```

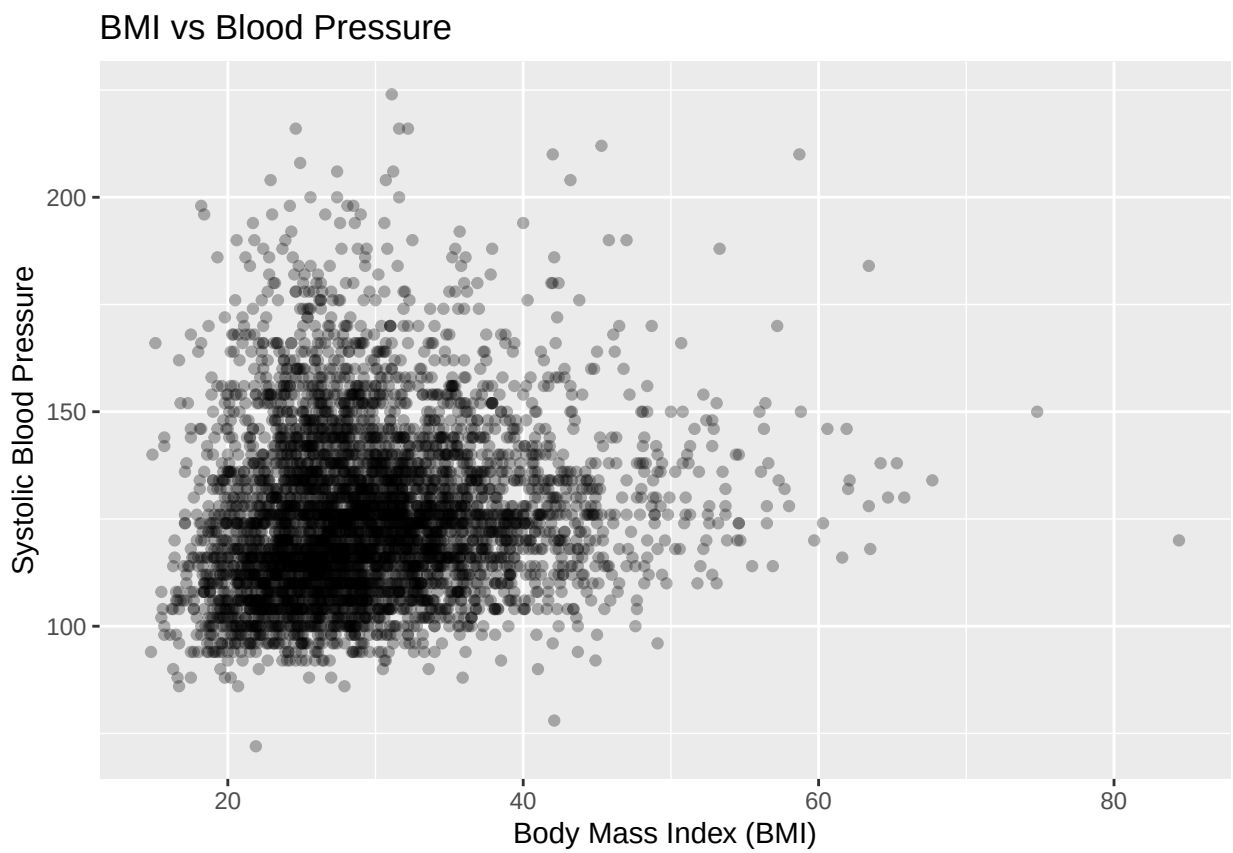
```r
df_clean <- df_clean %>%
  rename(
    age = RIDAGEYR,
    bmi = BMXBMI,
    bp = BPXSY1
  )

df_clean
```

```
## # A tibble: 4,833 x 5
##     SEQN   age   bmi    bp smoking
##    <dbl> <dbl> <dbl> <dbl>   <dbl>
##  1 93706    18  21.5   112       0
##  2 93709    75  38.9   120       1
##  3 93711    56  21.3   108       0
##  4 93712    18  19.7   112       1
##  5 93713    67  23.5   104       1
##  6 93715    71  22.5   112       1
##  7 93716    61  30.7   120       1
##  8 93717    22  24.5   116       1
##  9 93718    45  22     128       0
## 10 93721    60  35.9   132       0
## # i 4,823 more rows
```

```r
write_csv(df_clean, "../data/processed/nhanes_clean.csv")
```

```r
ggplot(df_clean, aes(x = bmi, y = bp)) +
  geom_point(alpha = 0.3) +
  labs(
    title = "BMI vs Blood Pressure",
    x = "Body Mass Index (BMI)",
    y = "Systolic Blood Pressure"
  )
```

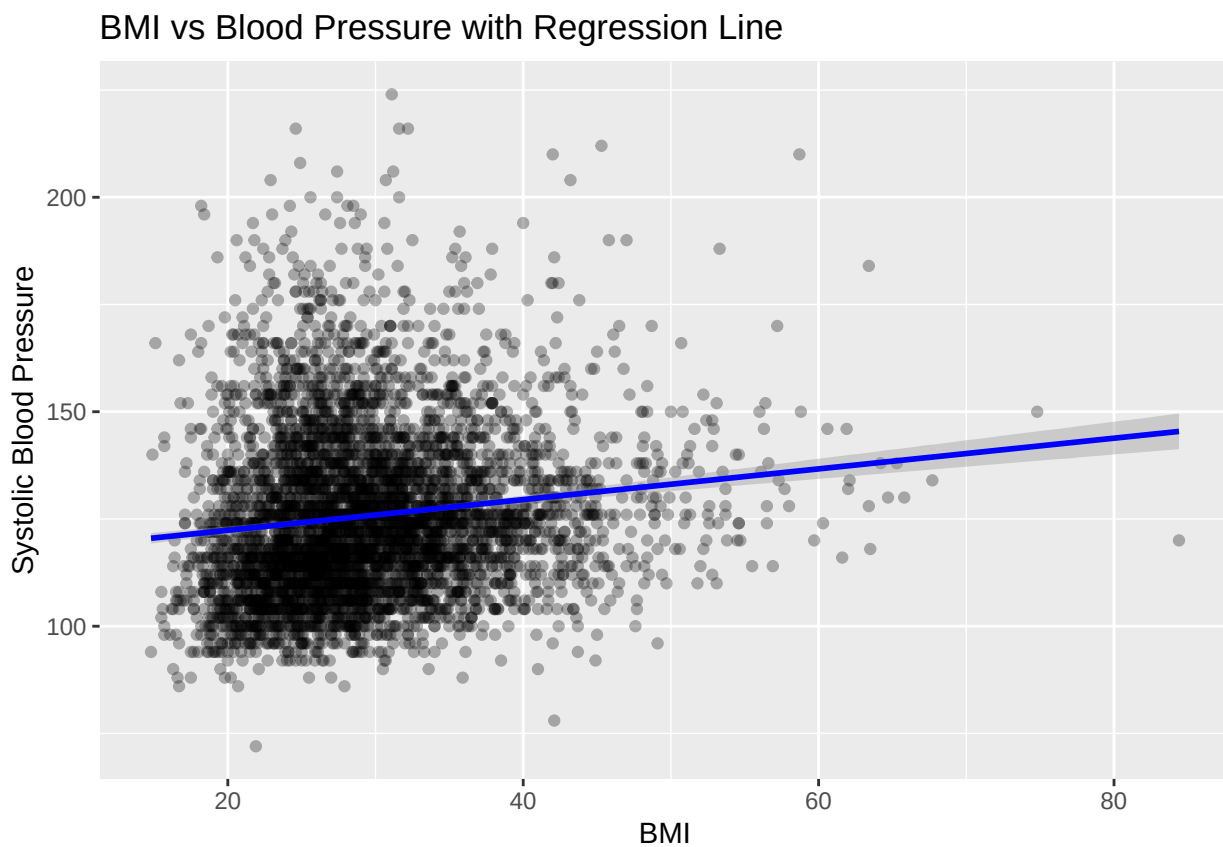


As BMI Increases –> Blood Pressure tends to increase

While BMI shows a positive association with systolic blood pressure, there is substantial variability, suggesting that additional factors (e.g., age, smoking) play an important role in explaining blood pressure differences.

```
ggplot(df_clean, aes(x = bmi, y = bp)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", color = "blue") +
  labs(
    title = "BMI vs Blood Pressure with Regression Line",
    x = "BMI",
    y = "Systolic Blood Pressure"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Model1: Simple Linear Regression Model

On average, how does blood pressure change as BMI changes?

$$bp = \beta_0 + \beta_1 bmi + \epsilon$$

Model Summary:

1. Coefficient for BMI A one-unit increase in BMI is associated with an average increase of about 0.36 units in systolic blood pressure.
2. Statistical significance: P-Value is very small There is very strong statistical evidence that BMI is positively associated with blood pressure.
3. R-squared BMI explains only about 1.8% of the variation in blood pressure.
4. Residual standard error On average, predictions are off by about ±19 units of blood pressure

While BMI shows a statistically significant positive association with blood pressure, it explains only a very small portion of the variation, suggesting that other factors play a substantial role.

```
model1 <- lm(bp ~ bmi, data = df_clean)

summary(model1)
```

```
##
## Call:
## lm(formula = bp ~ bmi, data = df_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -52.271 -13.472  -3.368   9.956  97.666
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 115.20504    1.15907  99.394   <2e-16 ***
## bmi           0.35786    0.03803   9.409   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.29 on 4831 degrees of freedom
## Multiple R-squared:  0.01799,    Adjusted R-squared:  0.01779
## F-statistic: 88.53 on 1 and 4831 DF,  p-value: < 2.2e-16
```

Model2: Multiple Linear Regression

$$bp = \beta_0 + \beta_1 bmi + \beta_2 age + \beta_3 smoking + \epsilon$$

Model2 Summary:

What is the relationship between BMI and blood pressure holding age and smoking constant?

1. BMI coefficient Holding age and smoking constant, a one-unit increase in BMI is associated with a 0.31 increase in blood pressure. Note: Part of the relationship in Model 1 was actually due to confounding (mainly age).
2. Age coefficient Each additional year of age is associated with about a 0.51 increase in blood pressure, holding BMI and smoking constant.
3. Smoking coefficient Smoking does not show a statistically significant association with blood pressure in this model.
4. R-squared The model now explains a much larger portion of the variation in blood pressure.
5. Residual standard error Improved from 19.3 to 16.8, hence, Predictions are more accurate now.

Conclusion:

The simple regression model shows a statistically significant positive relationship between BMI and systolic blood pressure. However, the model explains only a small fraction of the variation in blood pressure.

After including additional variables such as age and smoking status, the explanatory power of the model increases substantially. Age emerges as a strong and significant predictor, while the effect of BMI remains positive but slightly reduced, suggesting that part of the initial association was influenced by confounding factors.

Smoking does not appear to have a statistically significant relationship with blood pressure in this specification.

```r
model2 <- lm(bp ~ bmi + age + smoking, data = df_clean)

summary(model2)
```

```
##
## Call:
## lm(formula = bp ~ bmi + age + smoking, data = df_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -53.057 -11.239  -1.406   8.857  95.040
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 91.26104    1.17868  77.427   <2e-16 ***
## bmi          0.31268    0.03313   9.439   <2e-16 ***
## age          0.51326    0.01316  38.990   <2e-16 ***
## smoking     -0.25453    0.49822  -0.511    0.609
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.78 on 4829 degrees of freedom
## Multiple R-squared:  0.2573, Adjusted R-squared:  0.2569
## F-statistic: 557.8 on 3 and 4829 DF,  p-value: < 2.2e-16
```

Regression Table

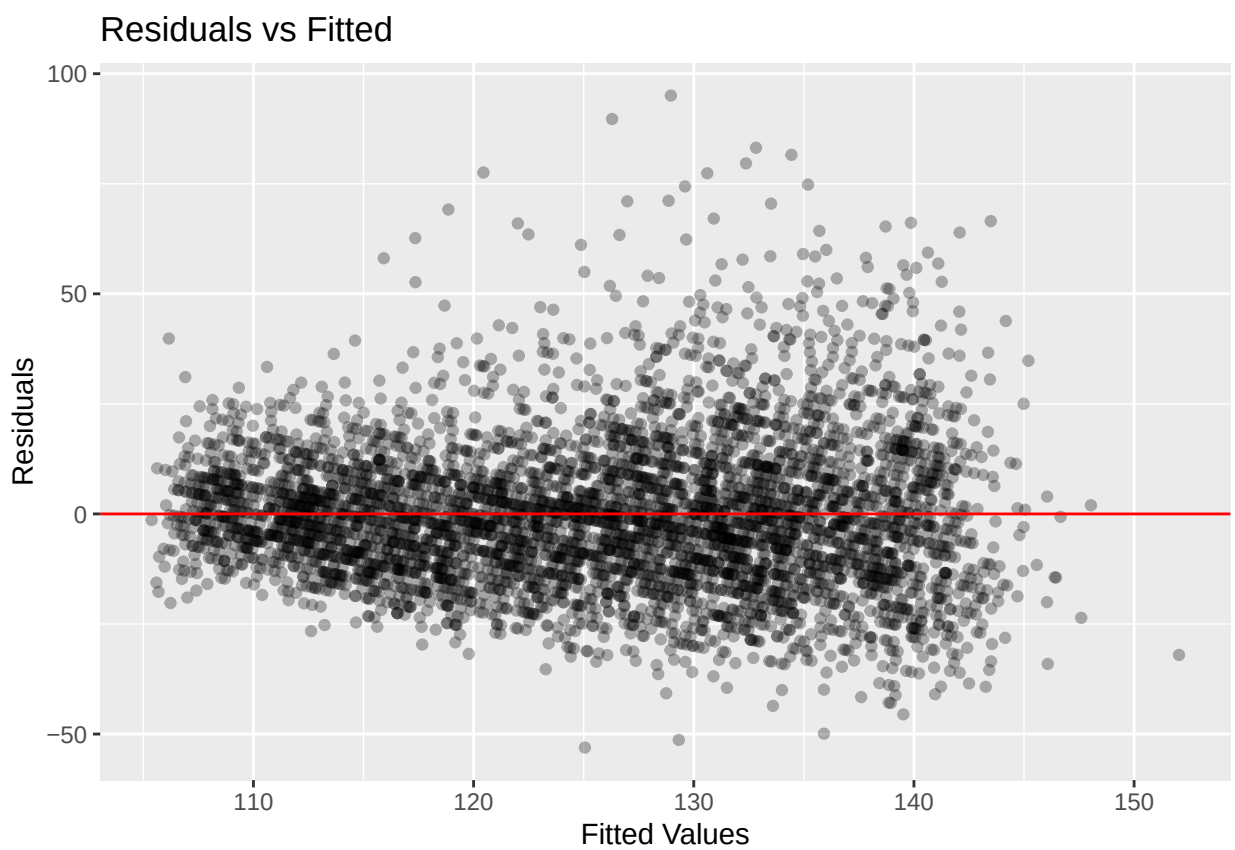
```r
stargazer(model1, model2,
          type = "text",
          title = "Regression Results: BMI and Blood Pressure",
          dep.var.labels = "Systolic Blood Pressure",
          covariate.labels = c("BMI", "Age", "Smoking"),
          digits = 3)
```

```
##
## Regression Results: BMI and Blood Pressure
## ==================================================================
##                                   Dependent variable:
##                       ---------------------------------------------
##                                Systolic Blood Pressure
##                              (1)                       (2)
## ------------------------------------------------------------------
## BMI                        0.358***                  0.313***
##                            (0.038)                   (0.033)
##
## Age                                                  0.513***
##                                                      (0.013)
##
## Smoking                                              -0.255
##                                                      (0.498)
##
```

```
## Constant                    115.205***                91.261***
##                              (1.159)                   (1.179)
##
## ---------------------------------------------------------------------
## Observations                  4,833                     4,833
## R2                            0.018                     0.257
## Adjusted R2                   0.018                     0.257
## Residual Std. Error    19.290 (df = 4831)       16.779 (df = 4829)
## F Statistic         88.527*** (df = 1; 4831) 557.791*** (df = 3; 4829)
## =====================================================================
## Note:                                       *p<0.1; **p<0.05; ***p<0.01
```

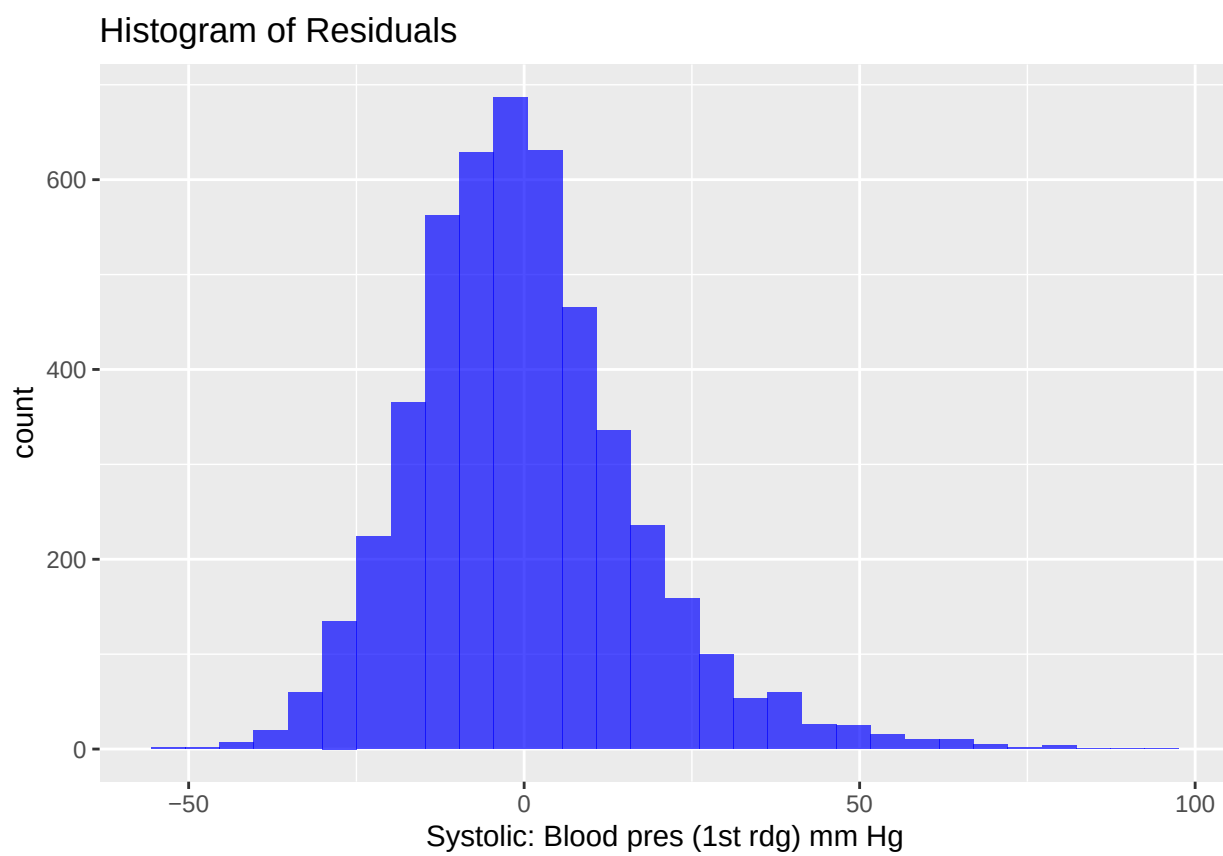
```r
residuals_model2 <- resid(model2)
fitted_model2 <- fitted(model2)
```

```r
ggplot(data.frame(fitted_model2, residuals_model2),
       aes(x = fitted_model2, y = residuals_model2)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs Fitted",
       x = "Fitted Values",
       y = "Residuals")
```



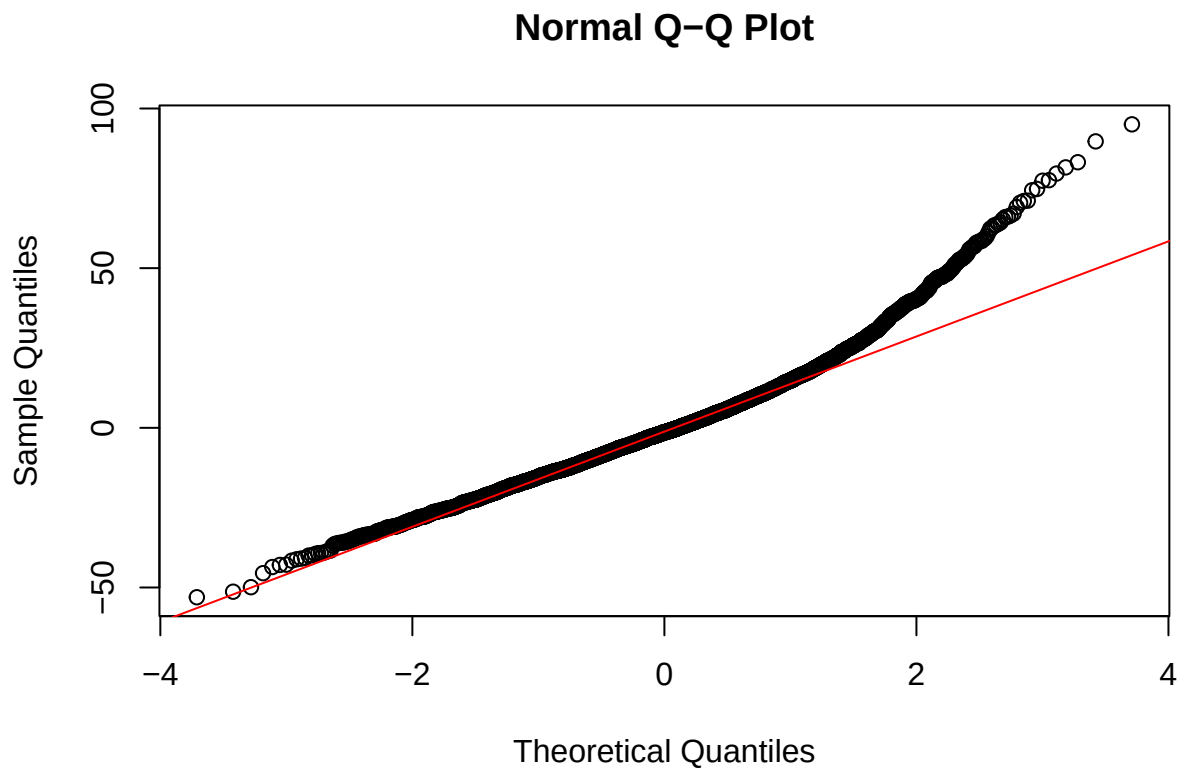
Histogram of Residulas

```
ggplot(data.frame(residuals_model2), aes(x = residuals_model2)) +
  geom_histogram(bins = 30, fill = "blue", alpha = 0.7) +
  labs(title = "Histogram of Residuals")
```



Q-Q Plot

```
qqnorm(residuals_model2)
qqline(residuals_model2, col = "red")
```



Regression Assumptions

The assumptions of the linear regression model were evaluated using residual diagnostic plots. The residuals versus fitted values plot shows that residuals are generally centered around zero without a clear systematic pattern, suggesting that the linearity assumption is reasonably satisfied. However, there is some evidence of increasing variability in residuals at higher fitted values, indicating mild heteroskedasticity.

The histogram of residuals shows an approximately bell-shaped distribution centered around zero, although with slight right skewness. This is further supported by the Q-Q plot, where most observations lie close to the reference line, with some deviation in the upper tail. These results suggest that the normality assumption is largely satisfied, with minor deviations at the extremes.

Overall, while there are small departures from ideal assumptions, particularly in terms of constant variance, these deviations are not severe and do not undermine the validity of the linear regression model for descriptive analysis.